

Midterm Project — Code Labs Briefs

Option B: Local Food Shelter & Volunteer Sign-up App

1. Project Overview & Scenario

Community food pantries and soup kitchens face unpredictable volunteer shift coverage and rapidly fluctuating inventory levels. You are tasked with creating a localized web application that displays real-time food pantry supply needs, allows volunteers to filter and register for available shifts, and provides immediate client-side validation to ensure sign-ups are recorded accurately.

2. Learning Objectives & Skill Alignment

- **HTML5:** Accessible forms, input attributes, custom attributes (data-*).
- **CSS3:** Flexible layout containers, CSS variables for theme management, responsive grid layouts.
- **Vanilla JavaScript:**
 - Custom form validation and regex handling.
 - Event handling (submit, change, input, click).
 - Dynamic DOM rendering from structured JavaScript objects/arrays.
- **Git & Web Hosting:** Deployment via Vercel or GitHub Pages.

3. Core Requirements & Features

Feature 1: Real-Time Pantry Status & Urgent Needs Dashboard

- **UI Structure:** Visual cards representing local food pantries, displaying current operation hours and critical shortage items (e.g., "Urgent Need: Baby Formula, Canned Soup").
- **Functionality:**
 - Include a status toggle (e.g., "Show Only Urgent Shortages") that filters pantries actively needing emergency donations.
 - Interactive counter showing total volunteers active today or total meals distributed (updatable via UI interaction).

Feature 2: Volunteer Shift Finder & Filter

- **UI Structure:** Listing of upcoming volunteer opportunities with details (Date, Time, Location, Role).
- **Functionality:**
 - Filter shifts by Shift Time (Morning, Afternoon, Evening) or Role (Food Sorting, Delivery Driver, Pantry Host).
 - Display a "Spots Remaining" badge that updates when a volunteer signs up.

Feature 3: Interactive Registration Form & Validation

- **UI Structure:** Sign-up modal or embedded form with fields: Full Name, Email, Phone, Emergency Contact, Preferred Shift, and Waiver Checkbox.
- **Functionality:**
 - Client-side validation: Prevent form submission if inputs are invalid (e.g., malformed email, missing required fields).
 - Display custom inline error messages next to invalid fields (no browser default tooltips only).
 - Show a confirmation modal/toast upon successful submission, updating the available spots on the corresponding shift card dynamically.

4. Technical Constraints & Submission

- **No External Frameworks:** Plain HTML, CSS, and Vanilla JS.

- **Deployment:** Must be hosted live on GitHub Pages or Vercel.
- **Video Walkthrough:** 3 to 5-minute video demonstrating shift filtering, form validation testing (valid and invalid inputs), and code organization.

5. Evaluation Rubric

Criteria	Weight	Requirements
HTML & Form Semantics	25%	Proper form elements (<label>, <input>, <select>, <button>), accessible markup.
CSS & Responsive Design	25%	Fluid layouts across all viewport sizes; clean CSS architecture (Flexbox/Grid).
JS Validation & DOM	35%	Custom JS validation, DOM manipulation on shift booking, filtering logic working flawlessly.
Git & Deployment	10%	GitHub repository with clear commit workflow, live deployed URL.
Video Walkthrough	5%	Professional demonstration of features and validation cases.